

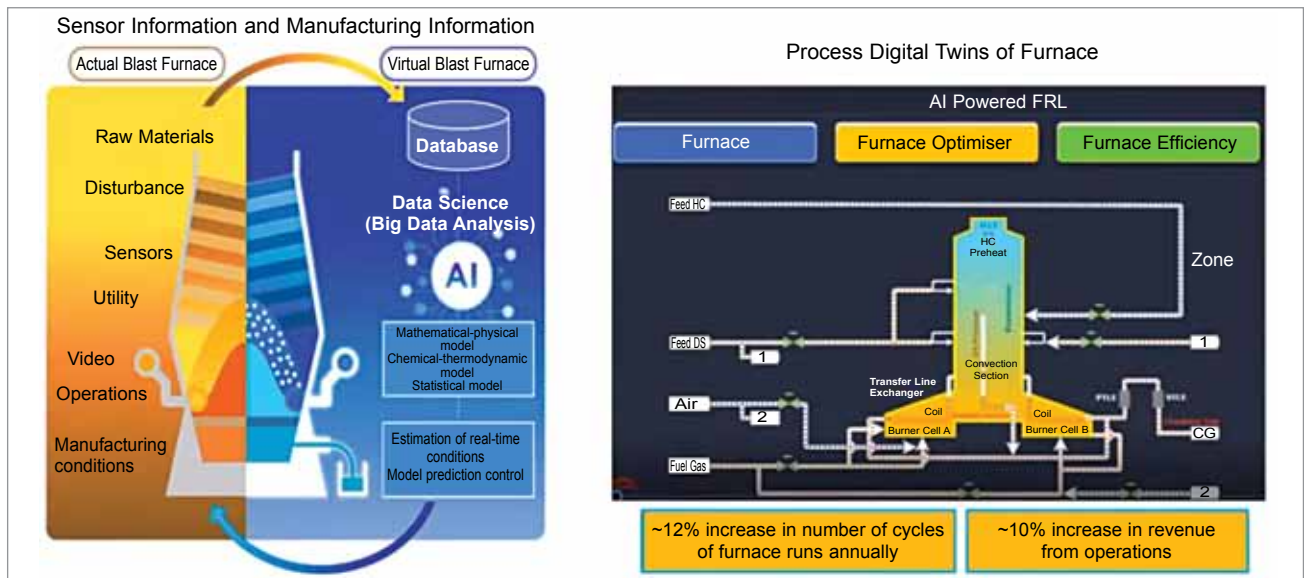


Fig. 2: Industrial furnace (Courtesy: LivSense)

Challenge of continuous process industry

The bigger challenge of this market is the low operational efficiency, which is affecting the earnings before interest, taxes, depreciation, and amortisation (EBITDA) by 4% to 7%. The next big challenge is the productivity performance, which is affected because of the complex and disconnected processes.

Various subsystems of the manufacturing plant do not bring the data to a central system for processing. Be it a predictive maintenance system for maintenance by original equipment manufacturers (OEMs) or other production processes controlled separately through a distributed control system (DCS) or historians, everything is worked on separate systems.

Over 90% of the data collected and stored in the plants is not being utilised properly for improving the performance of the plant, which may lead to an opportunity loss of \$167 billion. If the losses are being compared, the size of the petrochemical market is close to \$4,000 billion. This is the size of revenue being left on the table since the decision makers of the manufacturing plants are reluctant to take any chances of optimising the critical process parameters on a real-time basis.

Another challenge is a loss of the data silos. The productivity of employees is lost in processing information such as doing a root cause analysis as a post factor. This factor leads to a loss

in revenue by \$12.5 billion. The entire industry loses around \$200 billion, which is the reason for impacting the EBITDA by 4% to 7% directly.

In the current scenario, the manufacturing plants have a basic ecosystem in place with availability of PLCs, integrated machines, and sensors. There are also certain plants with automated systems for maintenance, production, quality, and inventory. However, all of these are separate and are not integrated. There exist advanced ARP systems which talk about logistics, product development, sales and marketing. With very little data being shared with adjacent systems, all of these systems operate separately.

The lack of reliable oriented maintenance for uptime maximisation, predictability of process variability, and overall equipment effectiveness (OEE) losses affect the whole value chain. These aspects are vital because real-time process deviations (process disturbances) are very difficult to capture, and none of the systems that are designed on first principles or on mathematical formulae based solutions can capture this accurately. Low accuracy leads to lower throughput and energy losses. Solutions to these problems can reduce OEE losses across the value chain and impact the overall yield of the plant.

These are the problems that need to be solved, and cognitive manufacturing promises to improve the

EBITDA by an average of 4-5%. The data from various systems has to be brought together to connect all the systems, and process it and operate with AI-led business processes, exhibiting the potential of AI in industries.

Addressing existing problems

The whole complex ecosystem of industries requires integration of various data systems. It is not just the sensor data system that needs retrofitting. As many systems are analogue, there exist multiple interfaces because of various proprietary and automation systems such as DCS, SCADA, Historian, and PLC systems. With multiple protocols a simplification of this ecosystem can be done by customisation, bringing data from all the heterogeneous processes to a big data platform, understanding the business processes and gaps, and applying the predictive and prescriptive analytics.

The way to implement this approach is to first use the traditional data that is already present in control systems. If no such systems are present, then sensors need to be retrofitted. The data should be brought to a big data platform, and with real-time AI and ML models, a combination of both thermodynamics and chemical engineering applied. The data can be visualised via built-in or custom widgets and flexible dashboards.

The key is to reimagine entire industries, be it petrochemical, chemi-